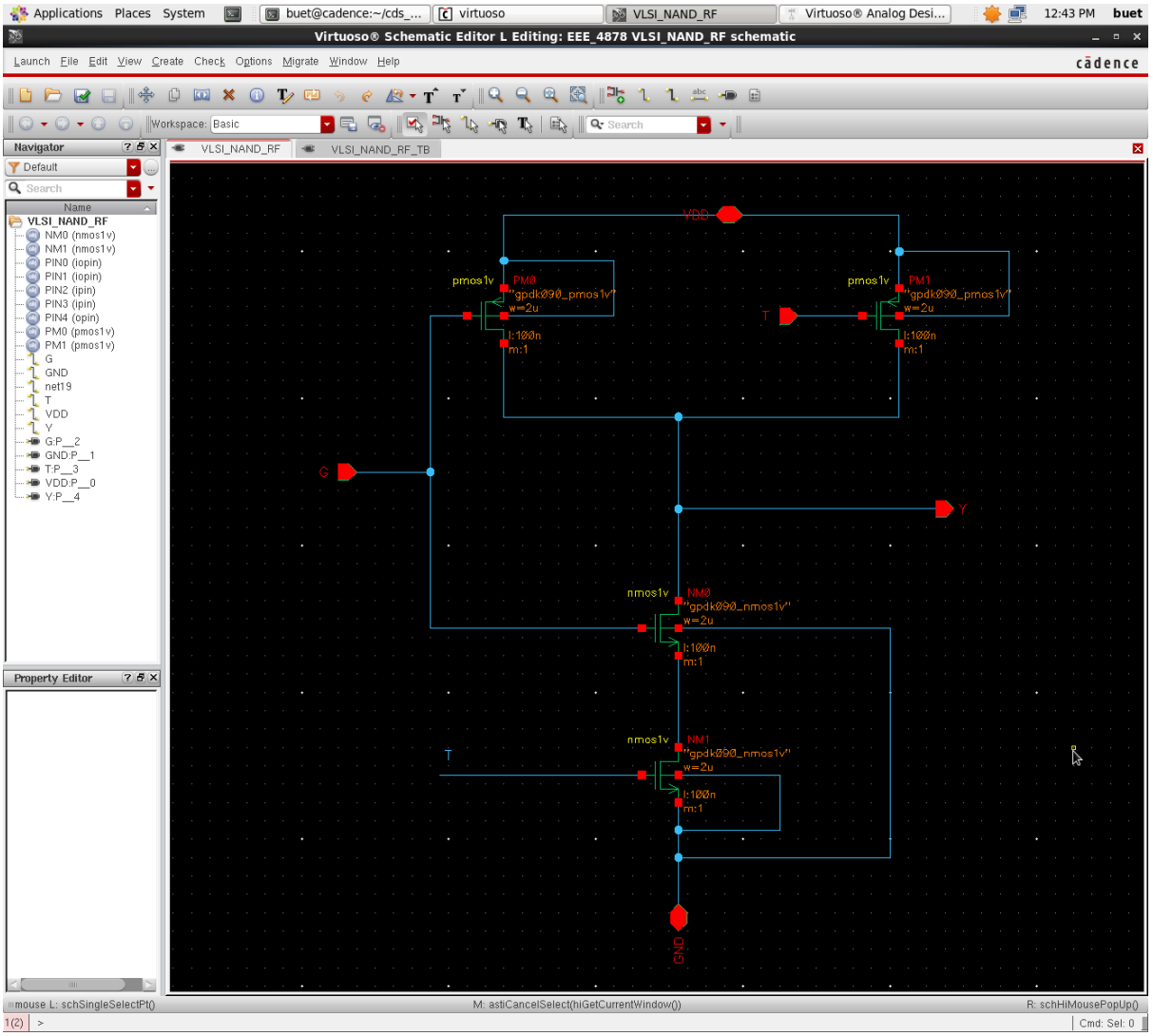
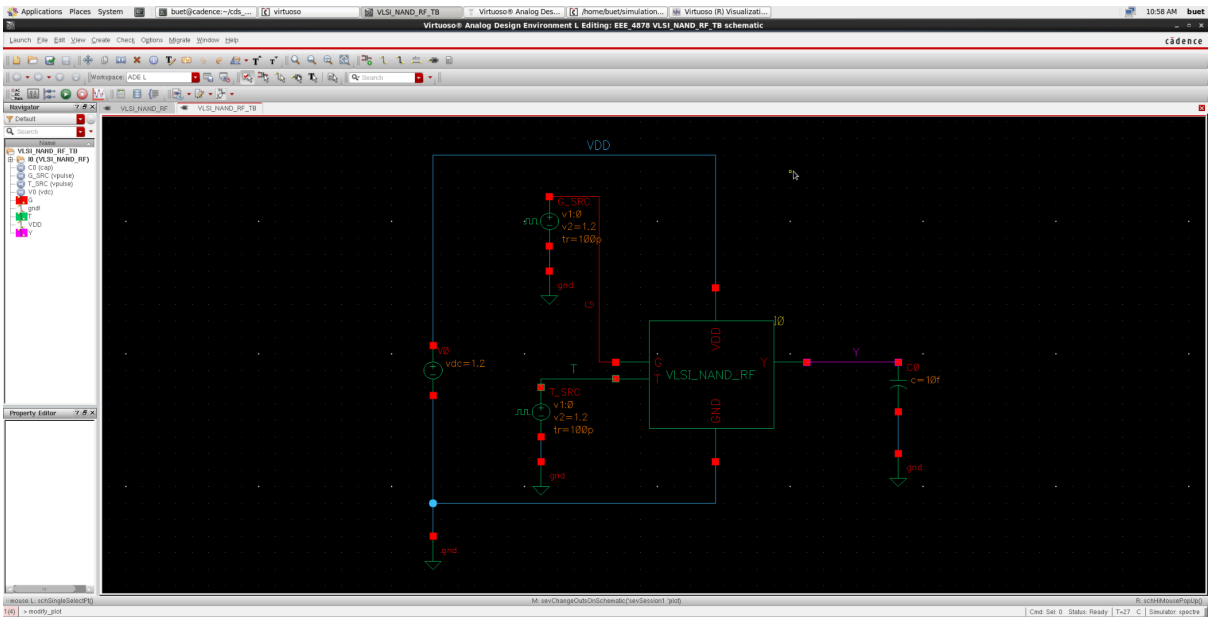
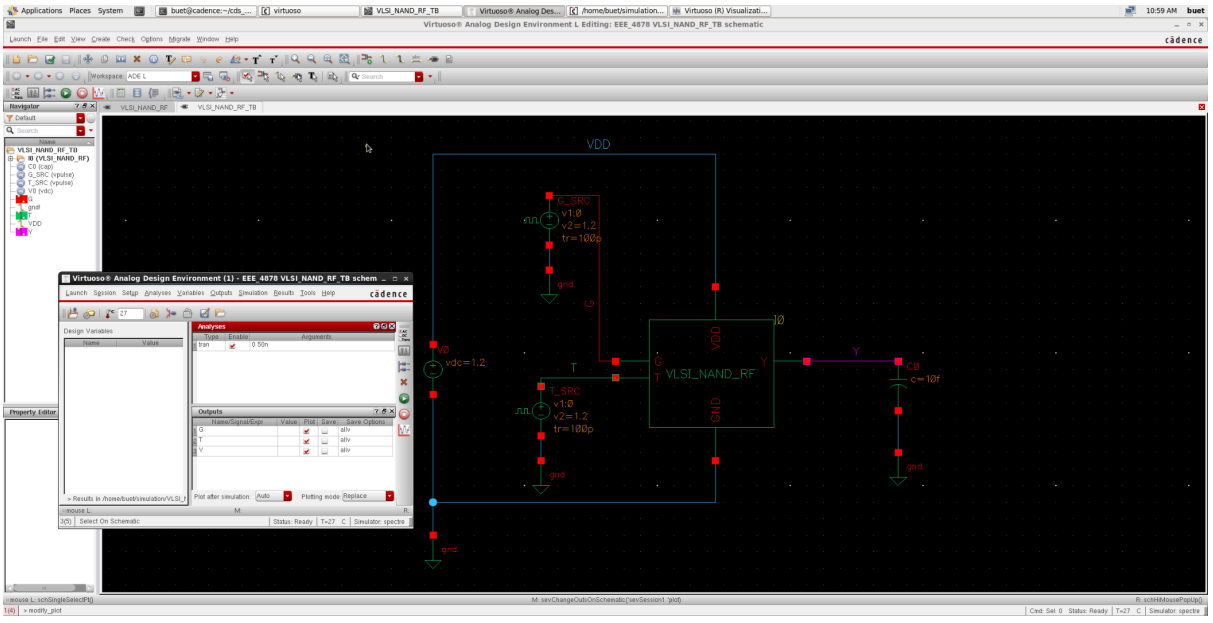
2 INPUT NAND GATE

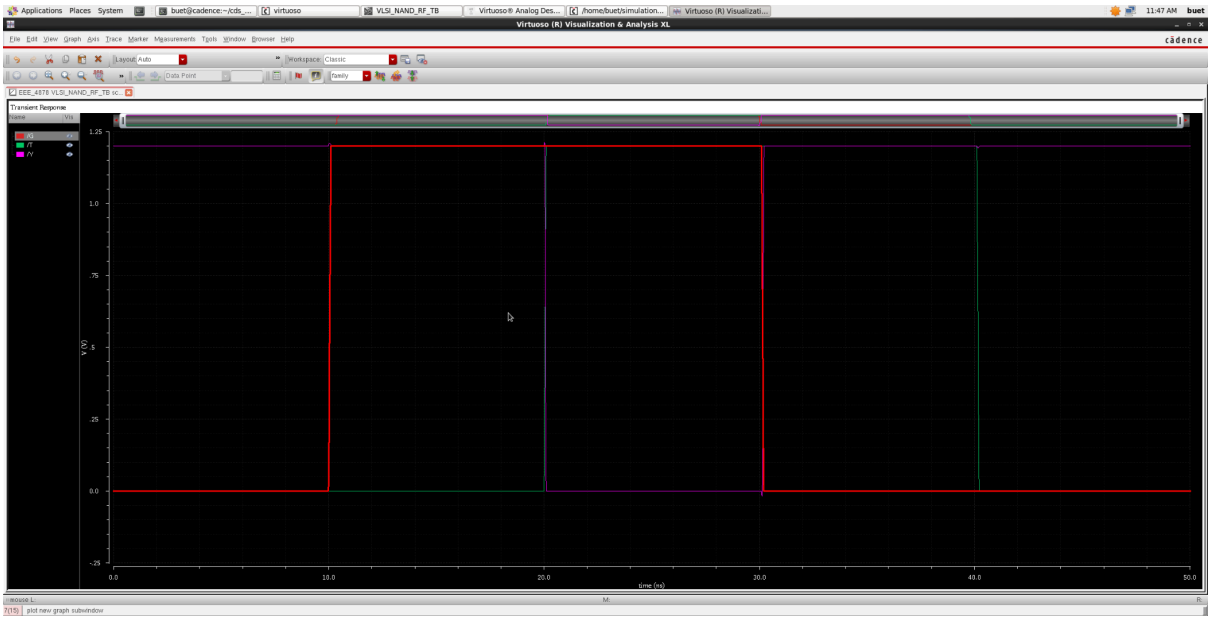


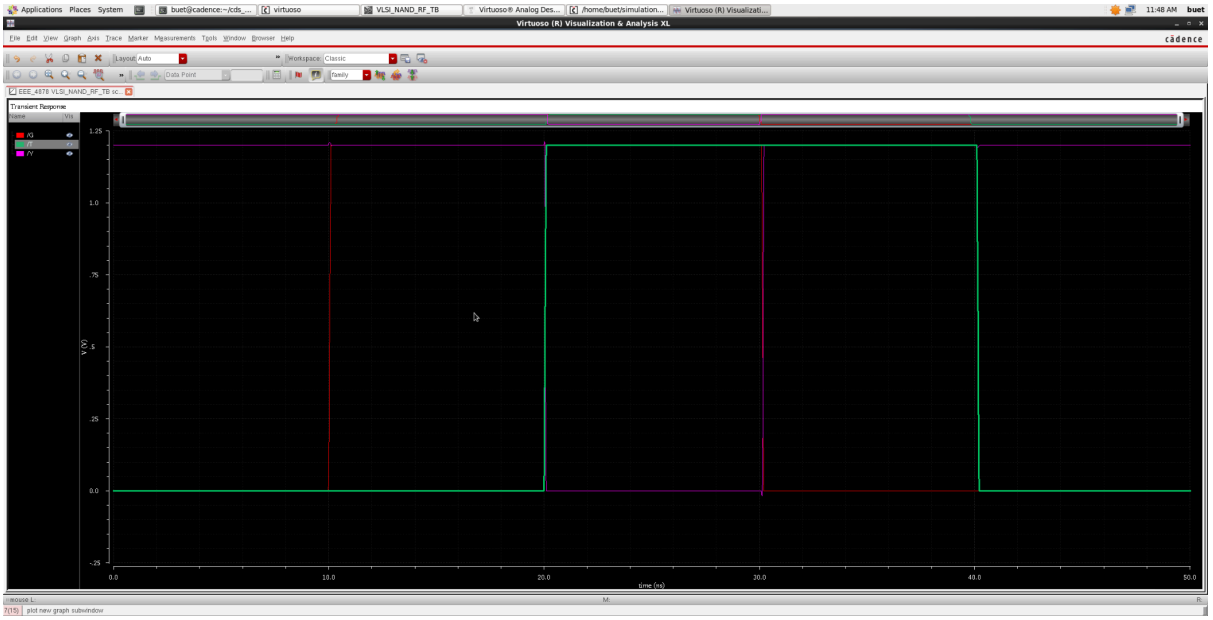
VLSI_NAND_RF_TB

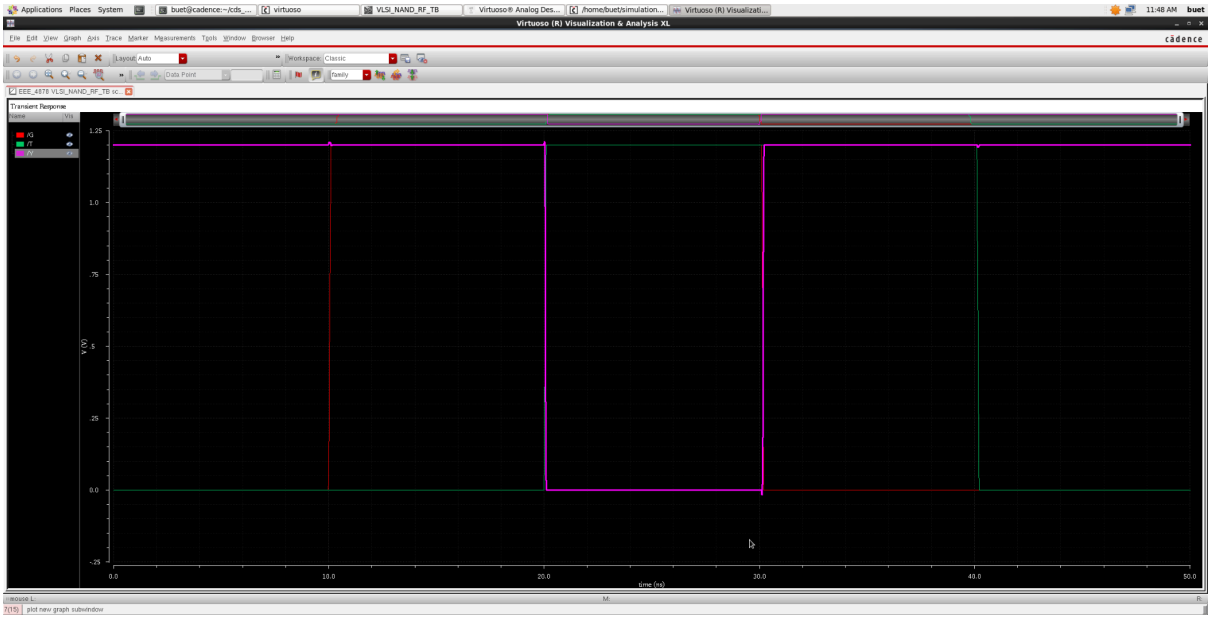




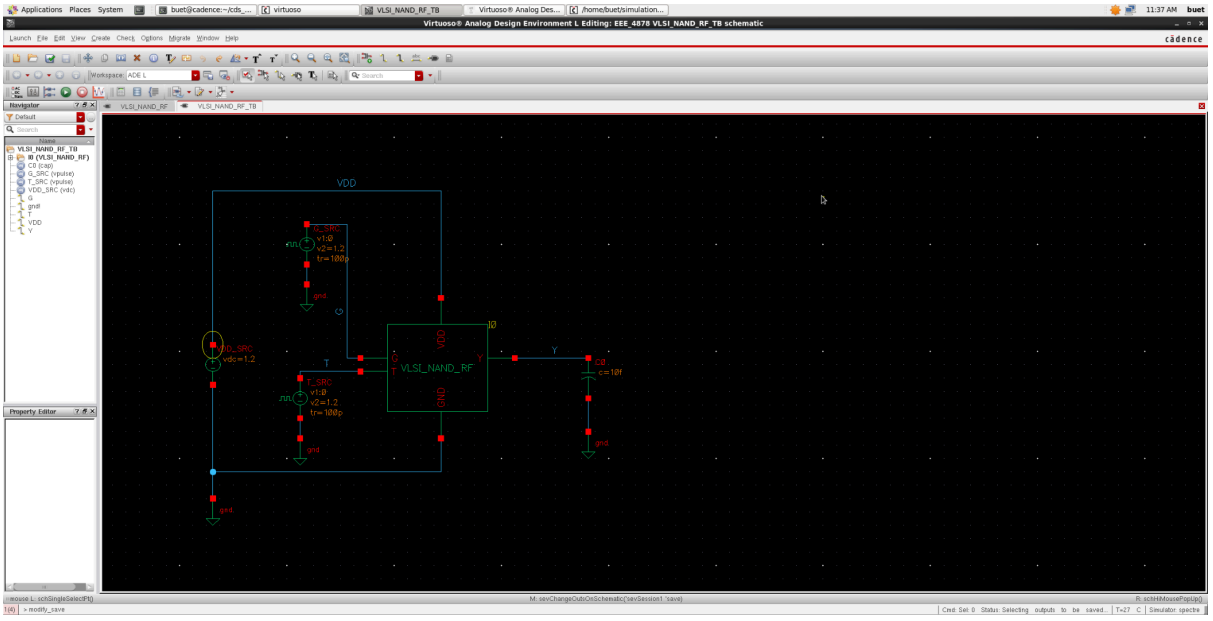
OUTPUTS


Virtuoso (R) Visualization & Analysis XL
Transient Response
V (V)
time (ns)

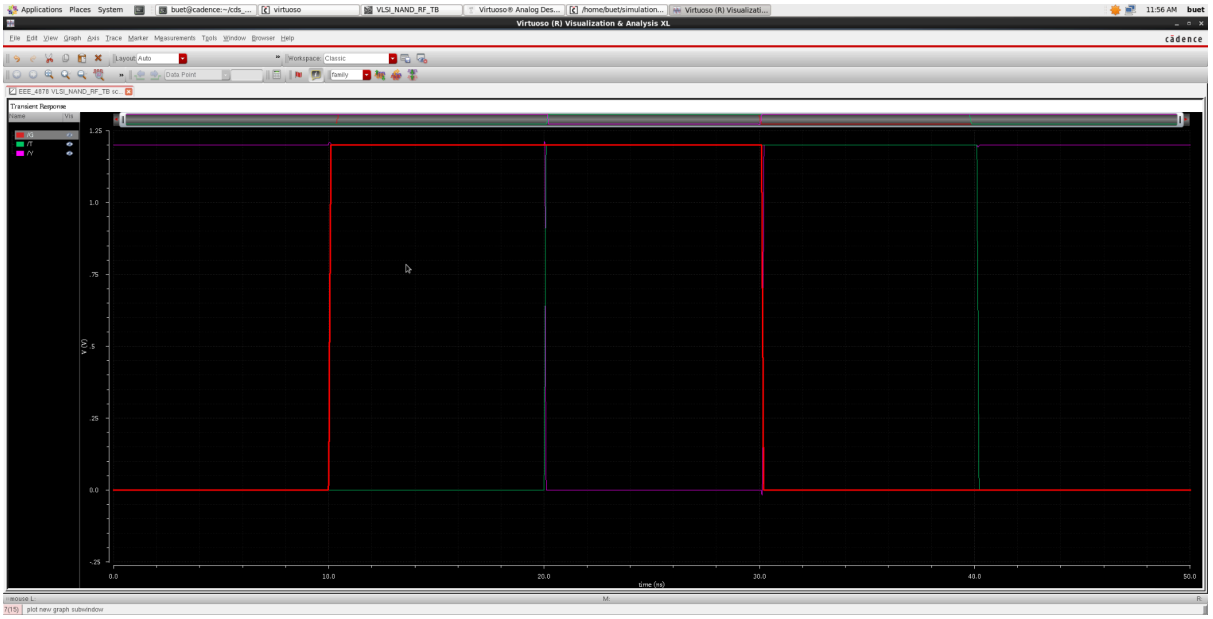

Virtuoso (R) Visualization & Analysis XL
Transient Response
V (V)
time (ns)

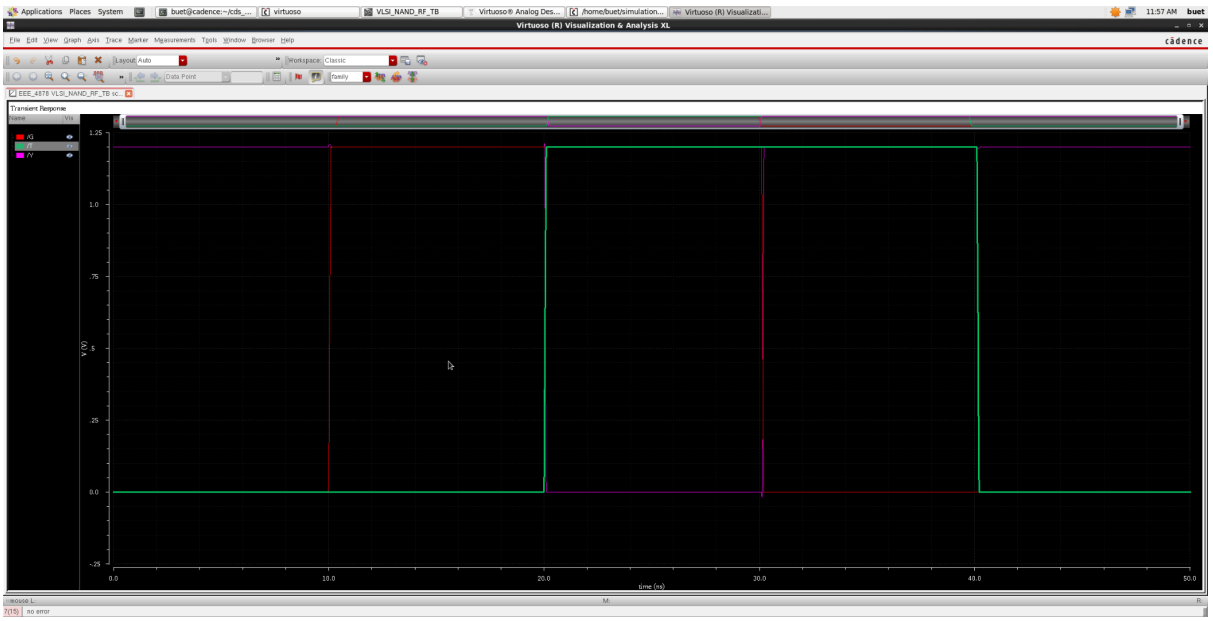


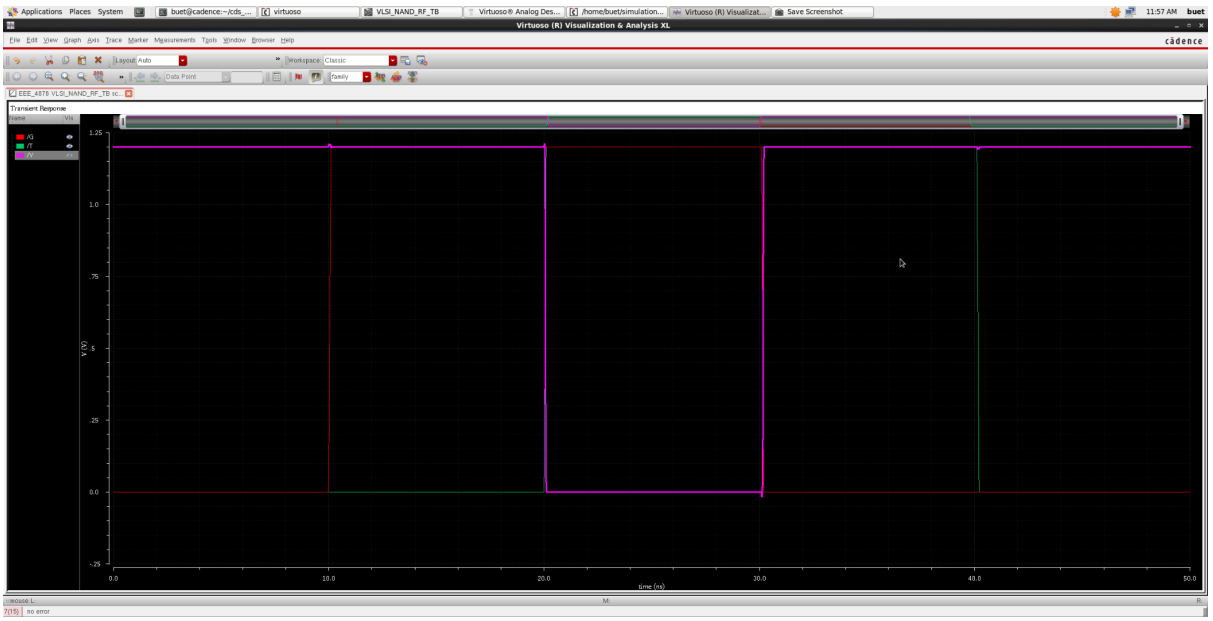
7. AVERAGE POWER MEASUREMENT



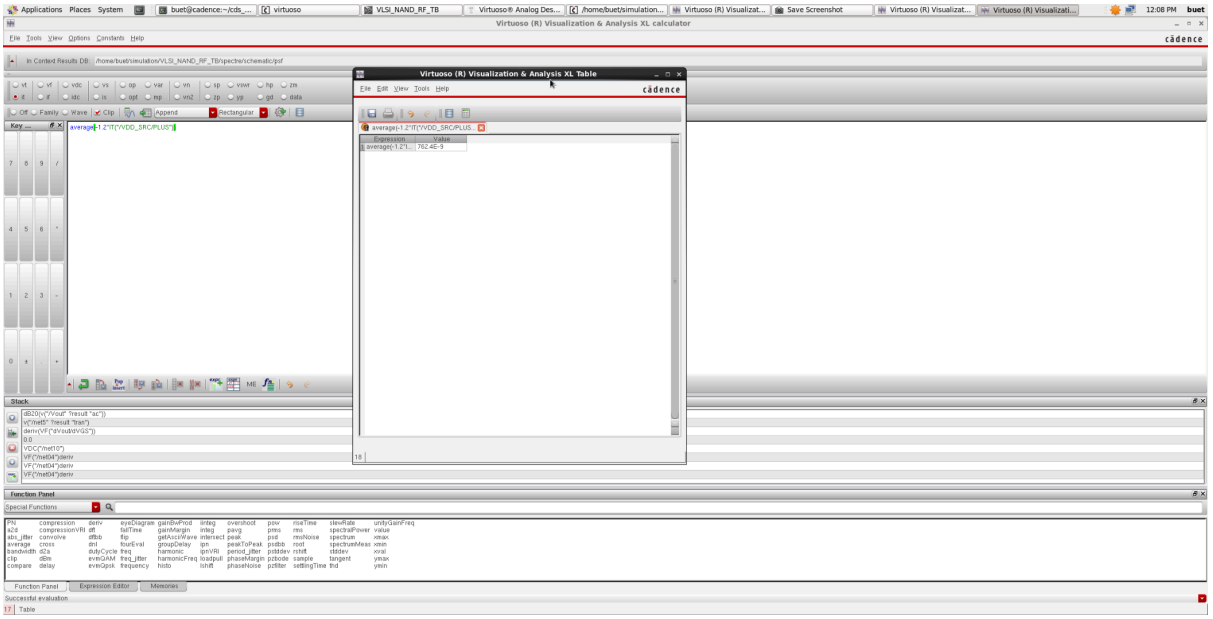
Running it by taking VDD_SRC (to be plotted)







Using calculator



The average-power result is:

Pavg =762.4×10^−9 W

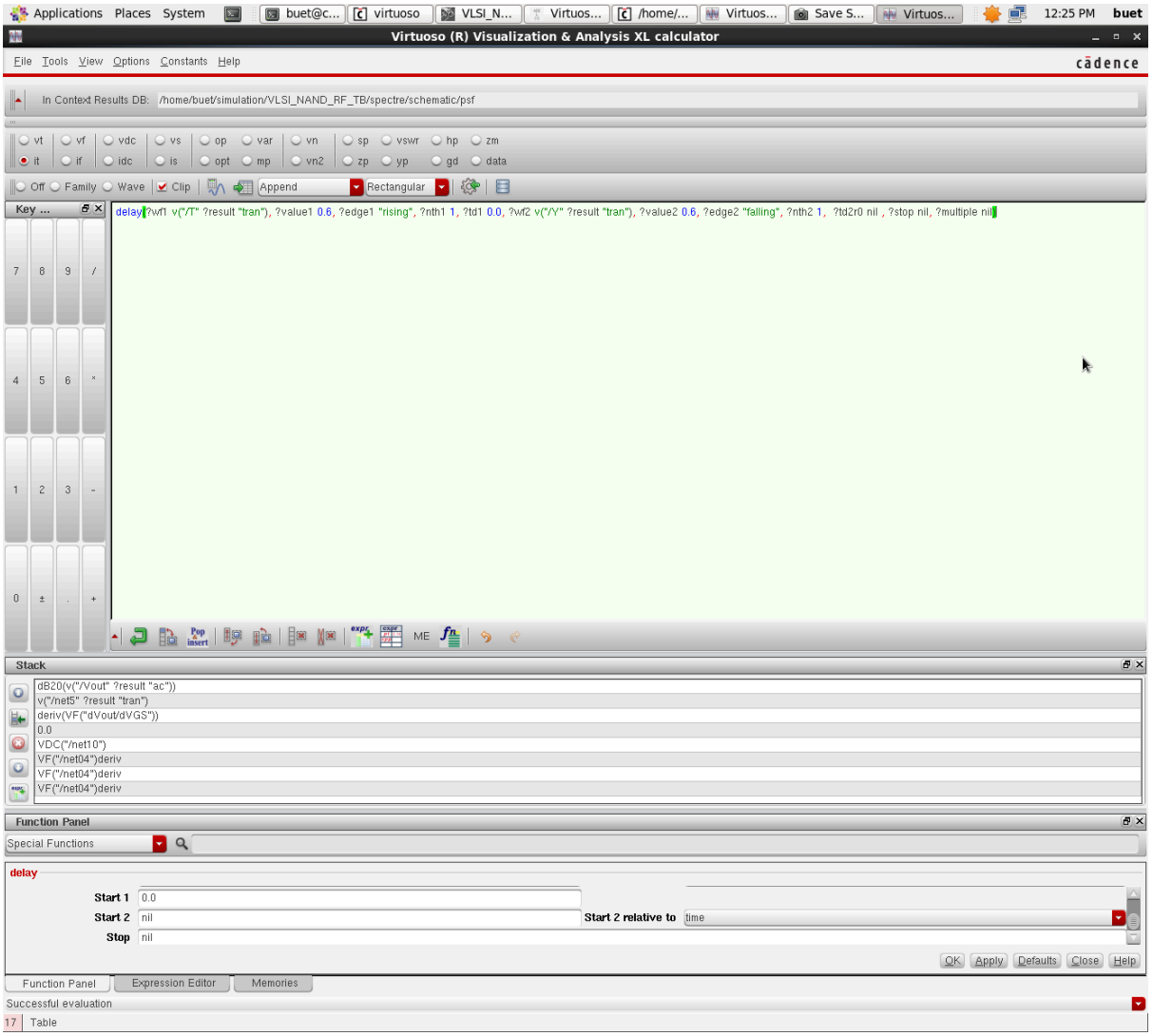
Therefore:

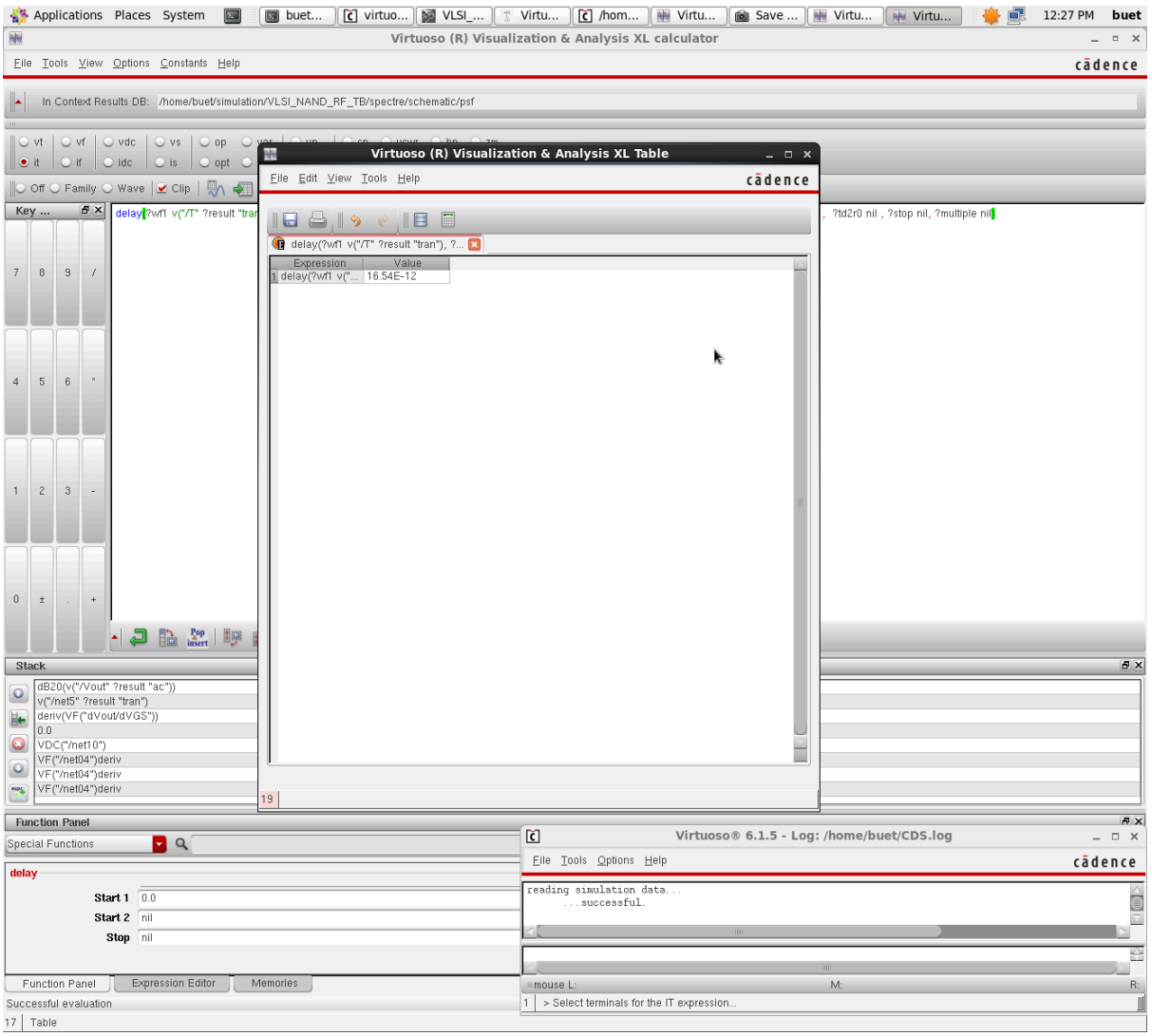
Pavg = 762.4 nW = 0.7624 μW

8 . PROPAGATION DELAY

It measures:

- T crossing 0.6 V while rising
- Y crossing 0.6 V while falling
- Difference between those crossing times = **tpHL**





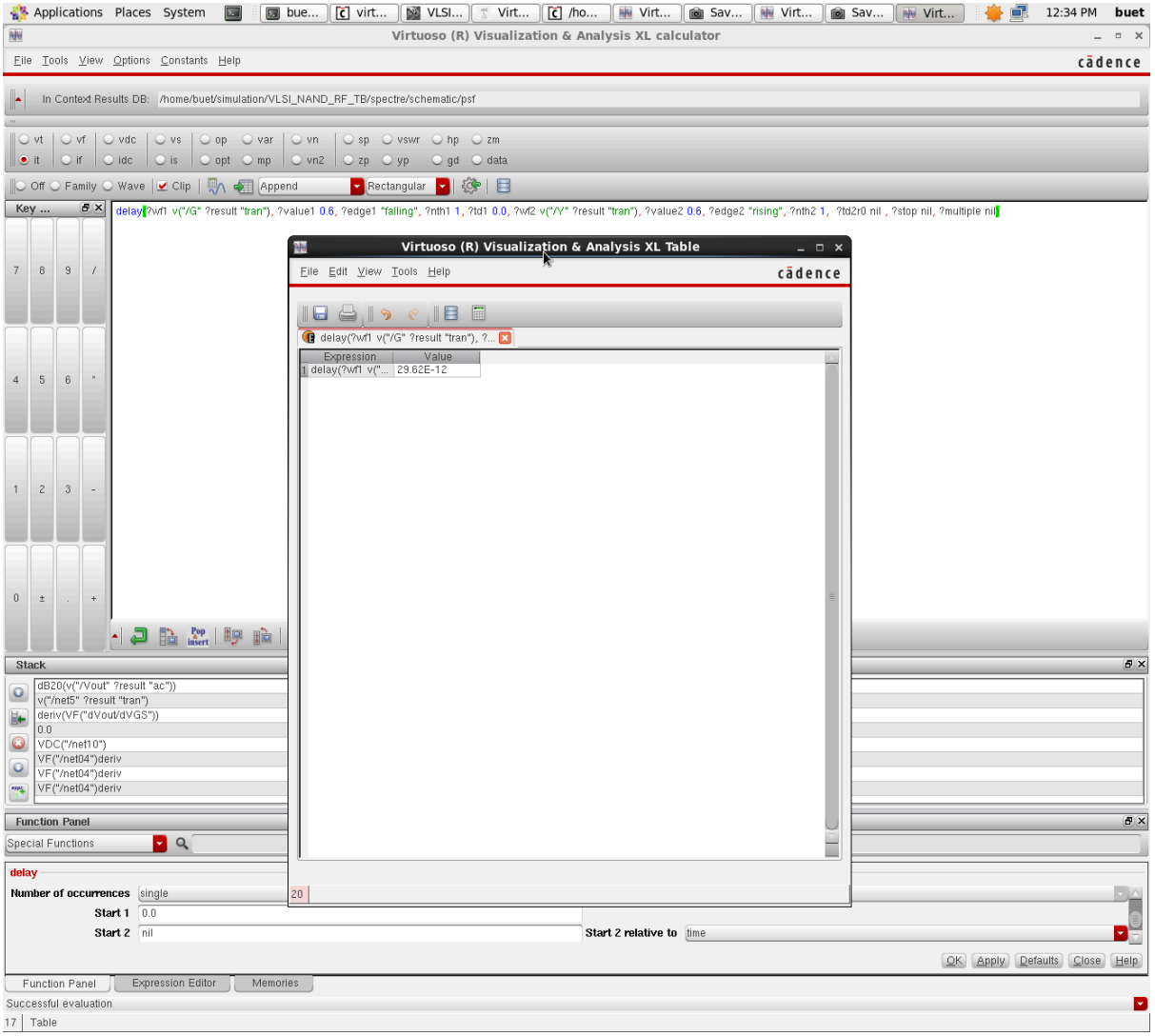
The result is:

tpHL = 16.54 × 10^{-12} s = 16.54 ps

This is the result for tpHL

Now

For tpLH



The result is:

- **tpLH = 29.62 ps**

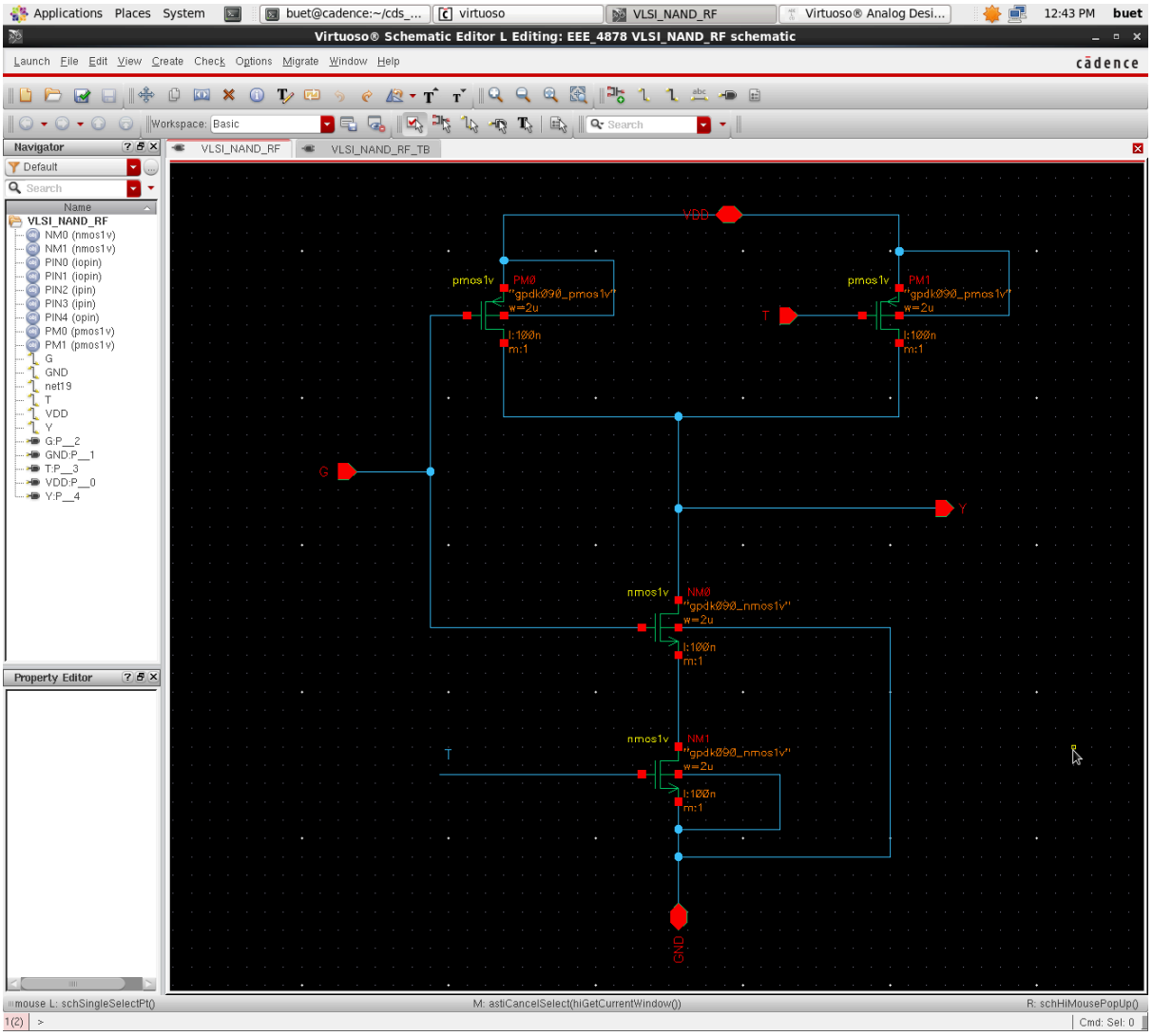
Average propagation delay:

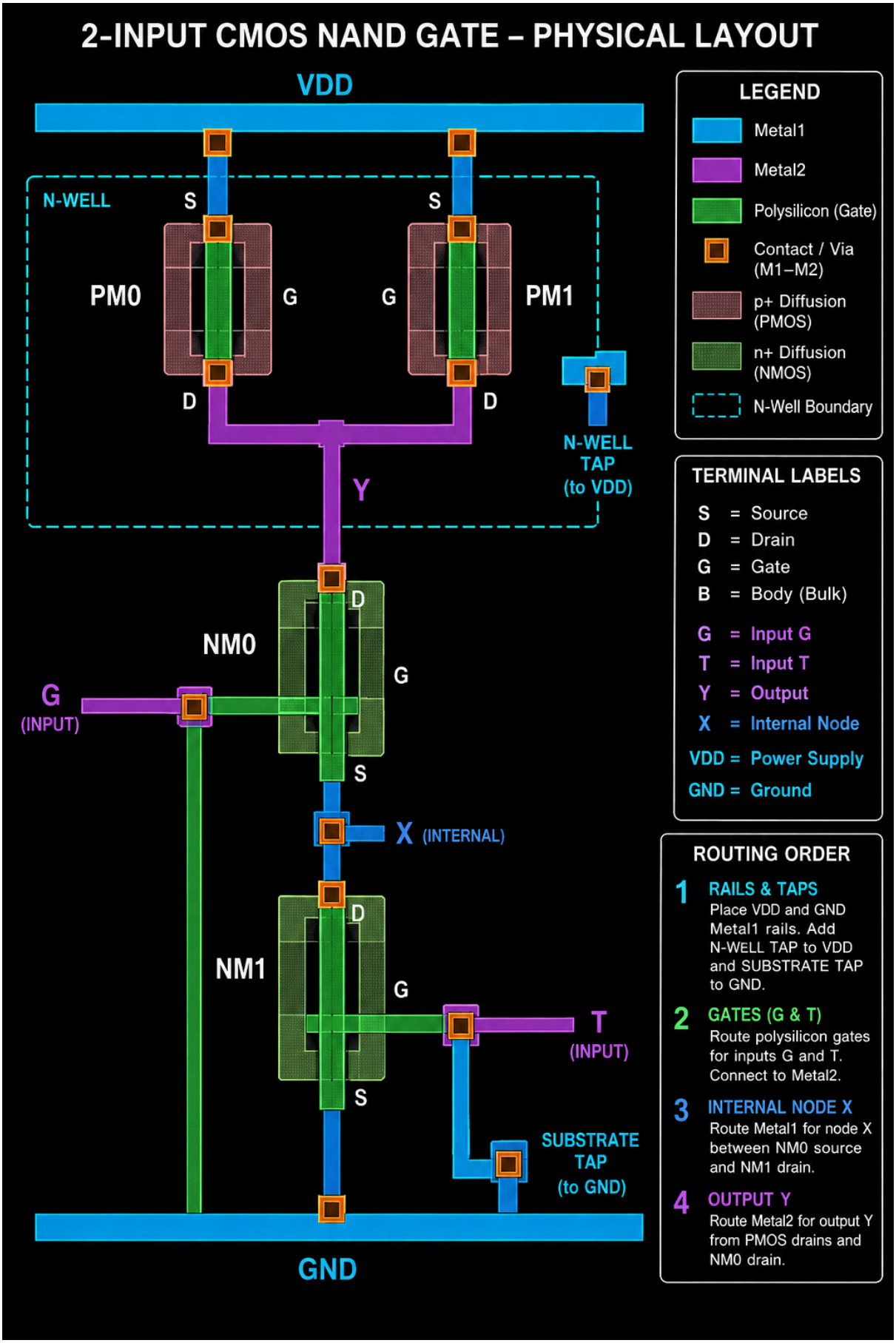
$$t_p = \frac{16.54 + 29.62}{2} = 23.08 \text{ ps}$$

completed results so far:

- **NAND logic operation: correct**
- **Average power: 762.4 nW**
- **tpHL: 16.54 ps**
- **tpLH: 29.62 ps**
- **Average delay: 23.08 ps**

9. Create the Physical Layout




2-INPUT CMOS NAND GATE – PHYSICAL LAYOUT
VDD
N-WELL
S
S
PM0
G
G
PM1
D
D
N-WELL TAP (to VDD)
Y
D
NM0
G
G (INPUT)
S
X (INTERNAL)
D
NM1
G
T (INPUT)
S
SUBSTRATE TAP (to GND)
GND
LEGEND
Metal1
Metal2
Polysilicon (Gate)
Contact / Via (M1–M2)
p+ Diffusion (PMOS)
n+ Diffusion (NMOS)
N-Well Boundary
TERMINAL LABELS
S = Source
D = Drain
G = Gate
B = Body (Bulk)
G = Input G
T = Input T
Y = Output
X = Internal Node
VDD = Power Supply
GND = Ground
ROUTING ORDER
1 RAILS & TAPS
Place VDD and GND Metal1 rails. Add N-WELL TAP to VDD and SUBSTRATE TAP to GND.
2 GATES (G & T)
Route polysilicon gates for inputs G and T. Connect to Metal2.
3 INTERNAL NODE X
Route Metal1 for node X between NM0 source and NM1 drain.
4 OUTPUT Y
Route Metal2 for output Y from PMOS drains and NM0 drain.